Lecture 21 (11/4)

Feature maps

Today:

1) Residual plots: the spike protein
2) Statistics and machine learning
3) Feature maps and polynomial regression
4) Why interpolation overfits

Example: the SARS-CoV-2 spike protein

Moulana et al. (2022) measured how strongly the spike protein binds the human ACE2 receptor for every combination of the 15 mutations in the Omicron BA.1 variant. Y is the log binding affinity of the spike protein to an enzyme ACE2. $X_i = 1$ if mutation i is present.

We use four sites, Q498R, N501Y, Y505H and K417N, and the model with all second-order interactions:

$$Y = \beta_0 + \sum_{i=1}^{4} \beta_i X_i + \sum_{i<k} J_{ik} X_i X_k + \epsilon.$$

This has $1 + 4 + 6 = 11$ coefficients. The other 11 mutations vary in the background and end up in ϵ.

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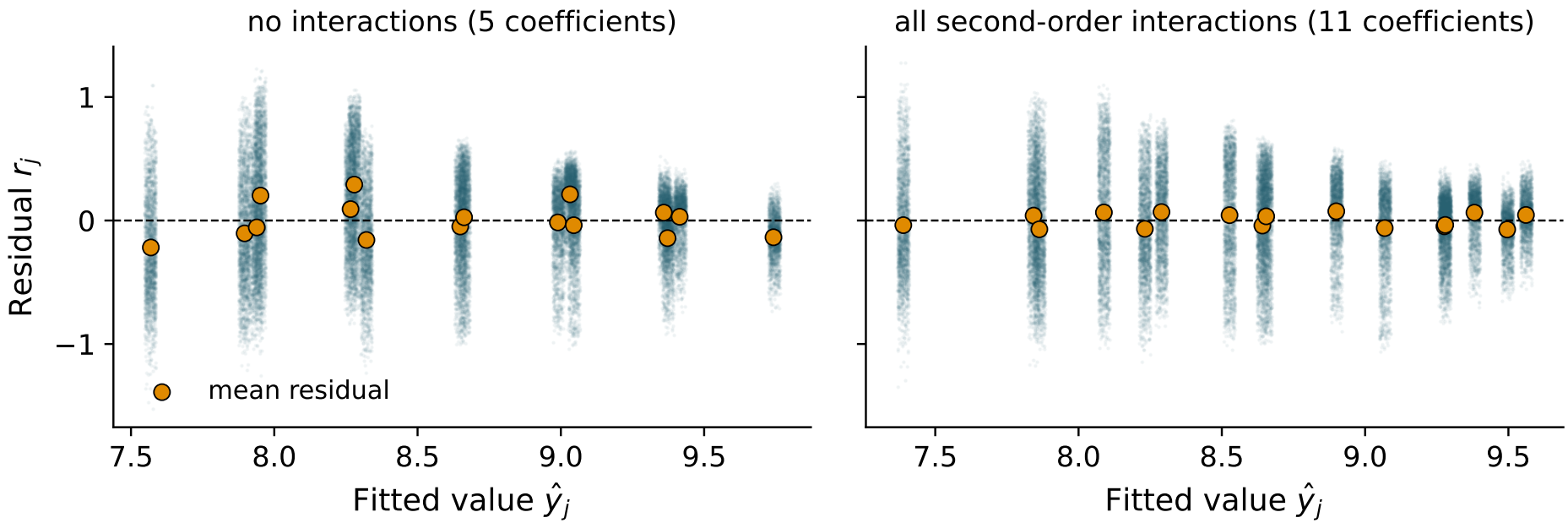
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$R^2 = 0.75$ without interactions (5 coefficients) and 0.78 with them.

Spike protein: residuals vs. fitted values

Each combination of the four sites has one fitted value, so the residuals fall in 16 vertical bands. The orange dots are the mean residual in each band.



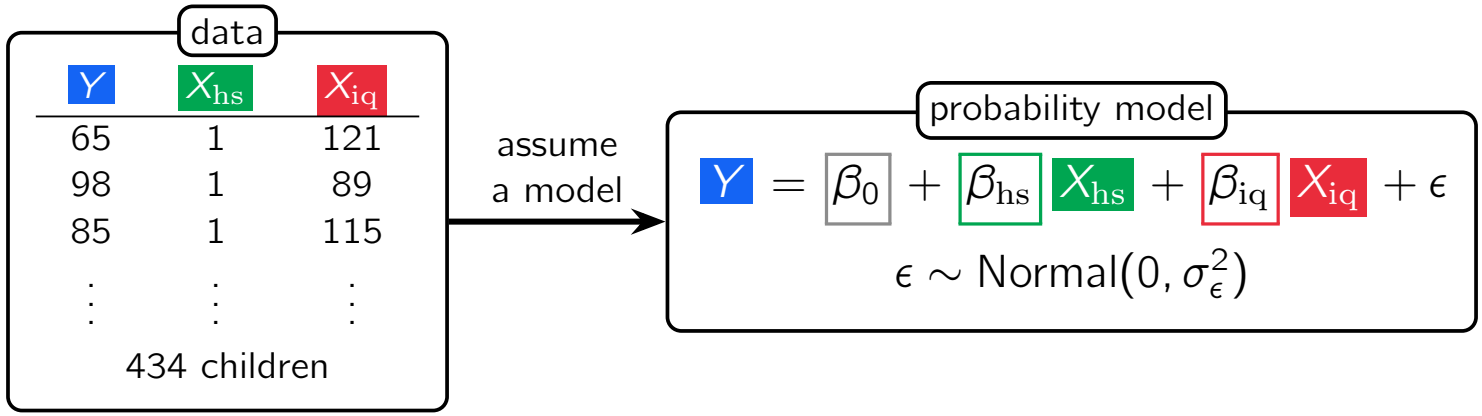
Statstics: Recap

data

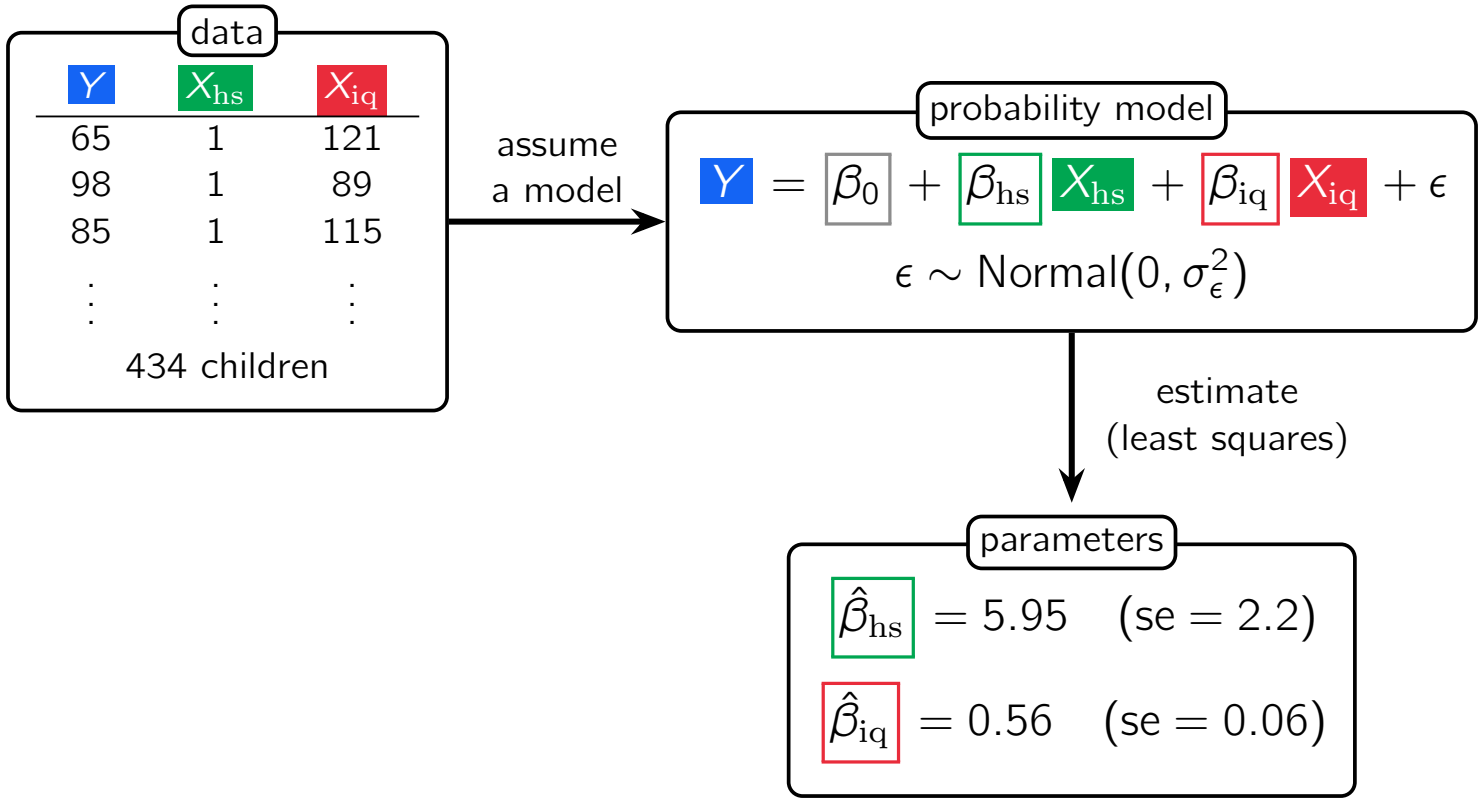
Y	X_{hs}	X_{iq}
65	1	121
98	1	89
85	1	115
⋮	⋮	⋮

434 children

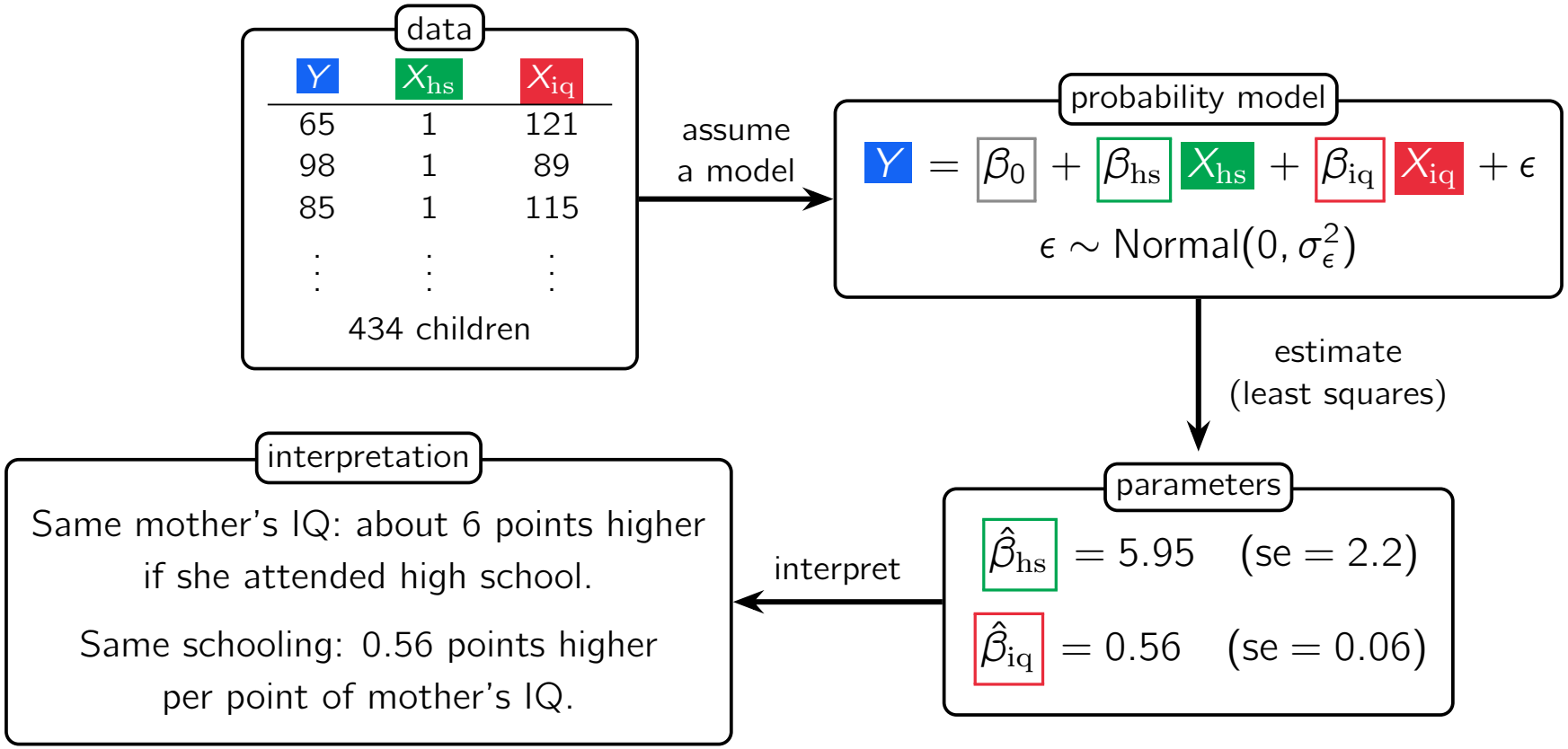
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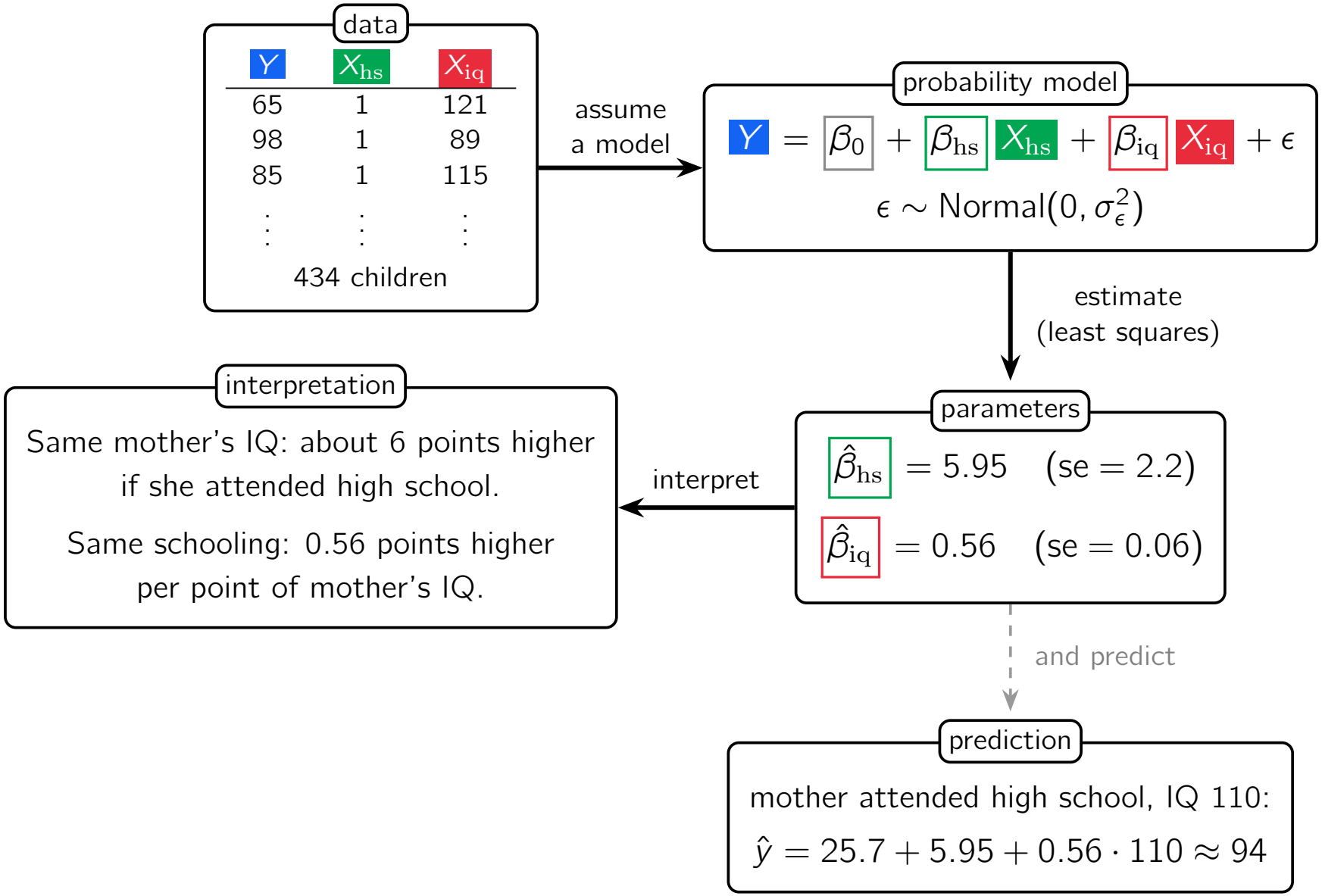
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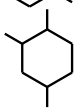


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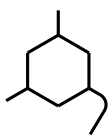


Example: drug discovery

data: N molecules

molecule	binding energy
	y_1
	y_2
	y_3
⋮	⋮
	y_N

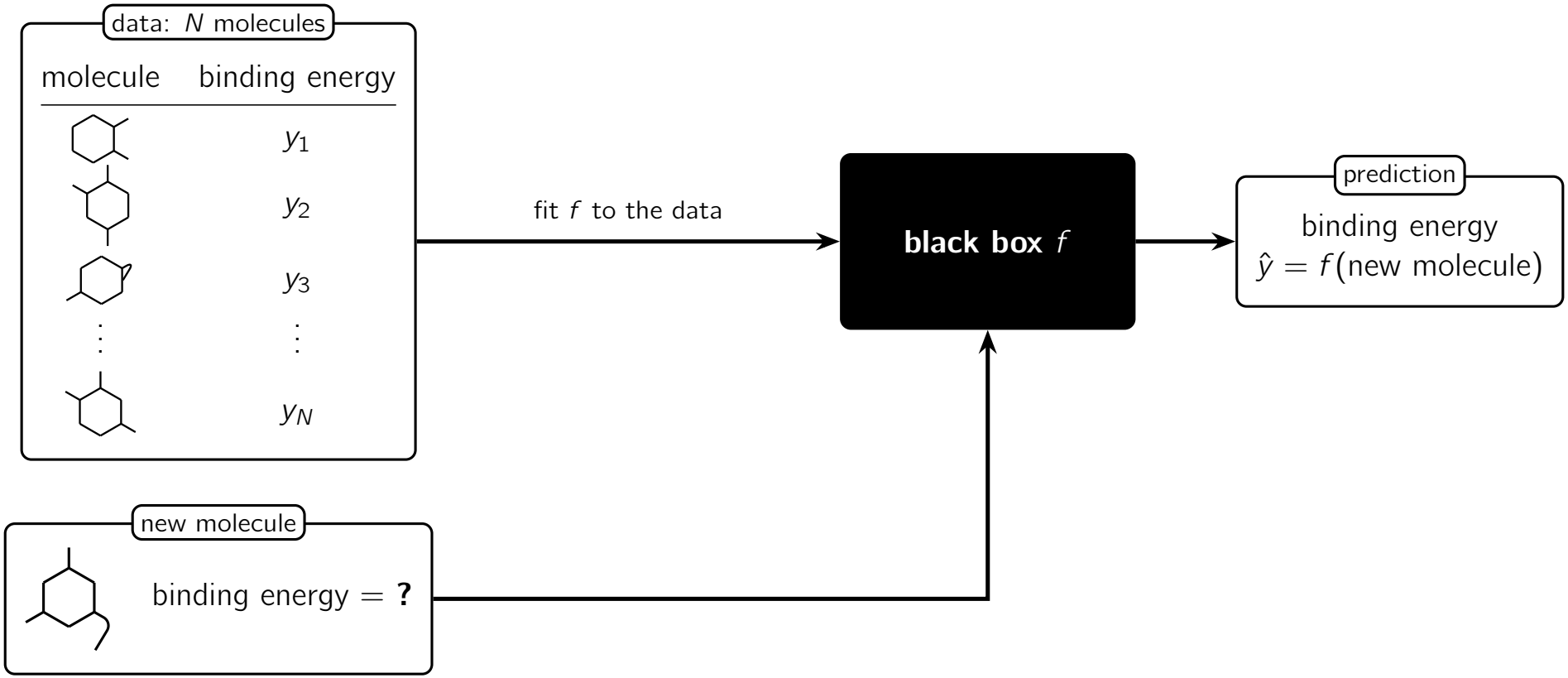
new molecule



binding energy = **?**

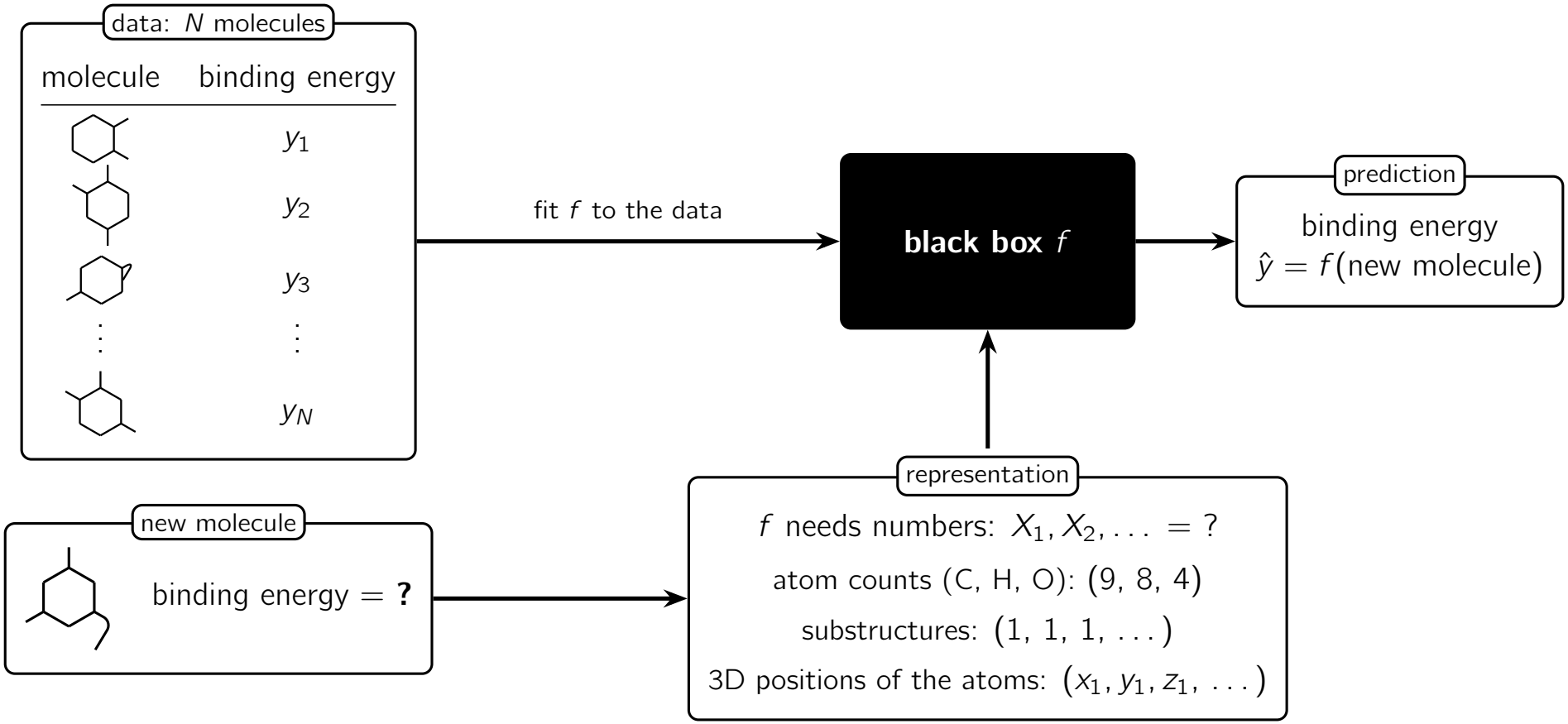
The input is a chemical structure which can be represented in different ways. The more complex input data and choice of representation makes this very different from the test scores and other examples.

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Further reading: L. Breiman (2001), Statistical modeling: the two cultures, *Statistical Science* 16(3), 199–231.

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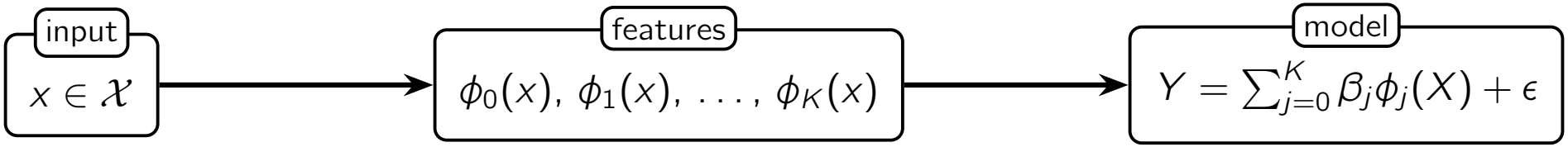
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What does it mean for a model to be complex? Does having more parameters make it more complex? Can a model be too complex?

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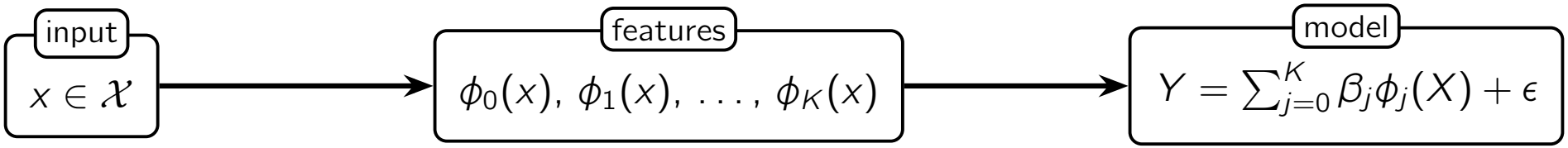
Learning a function

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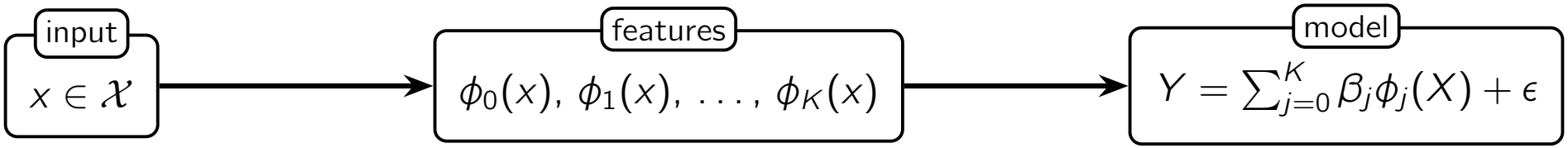
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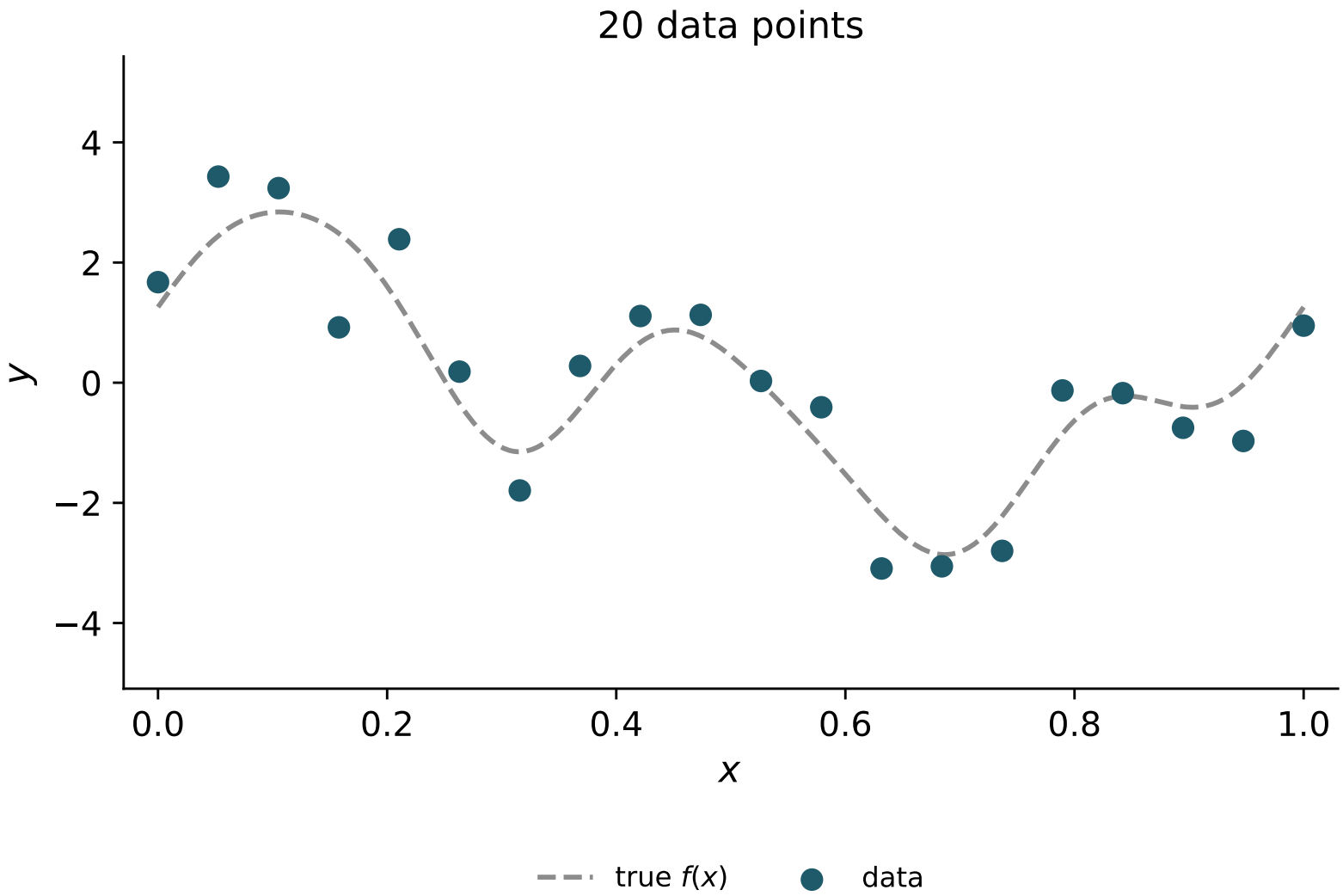


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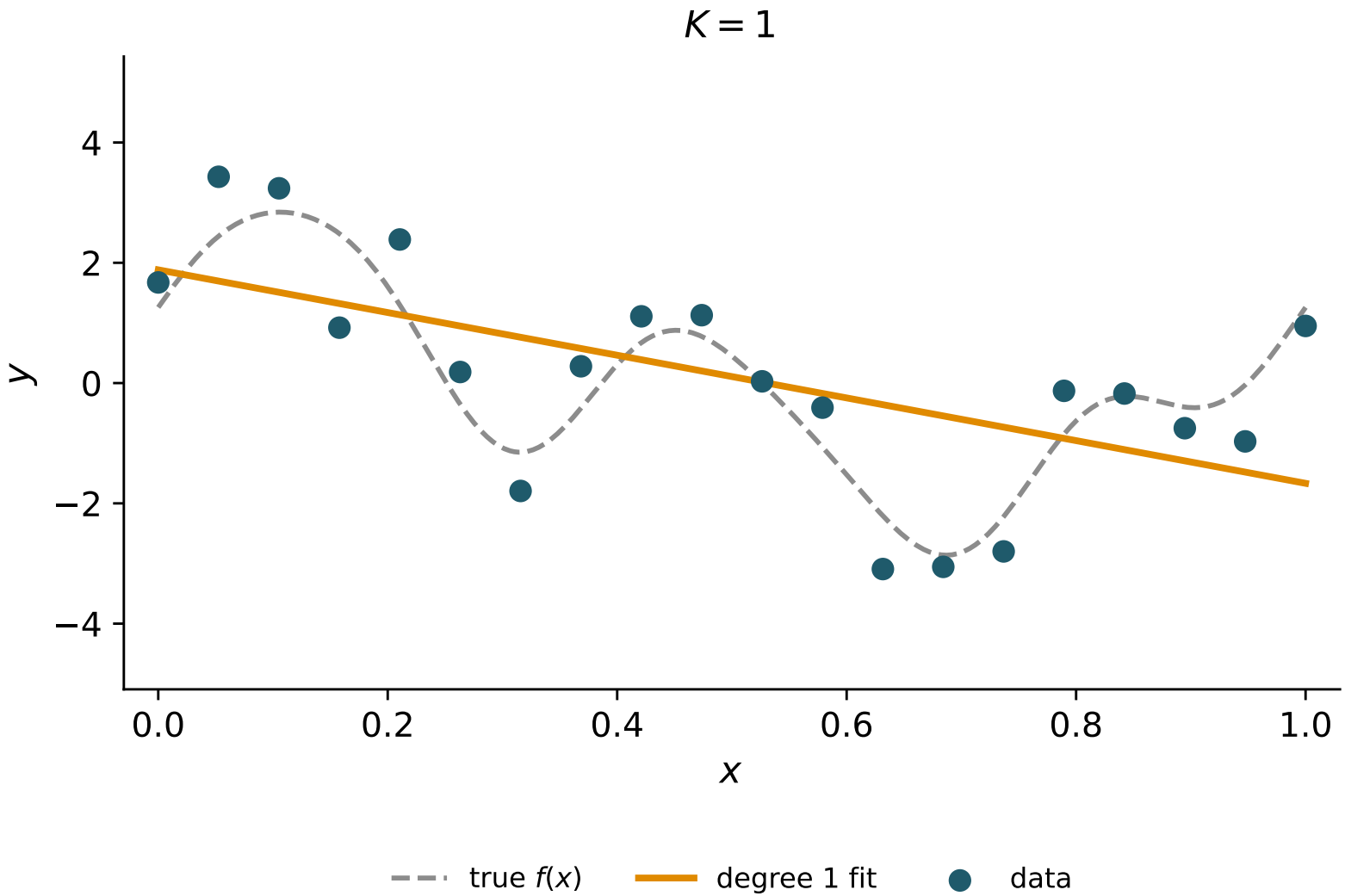
Polynomial regression: $\mathcal{X} = \mathbb{R}$ and $\phi_j(x) = x^j$, so

$$Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \cdots + \beta_K X^K + \epsilon.$$

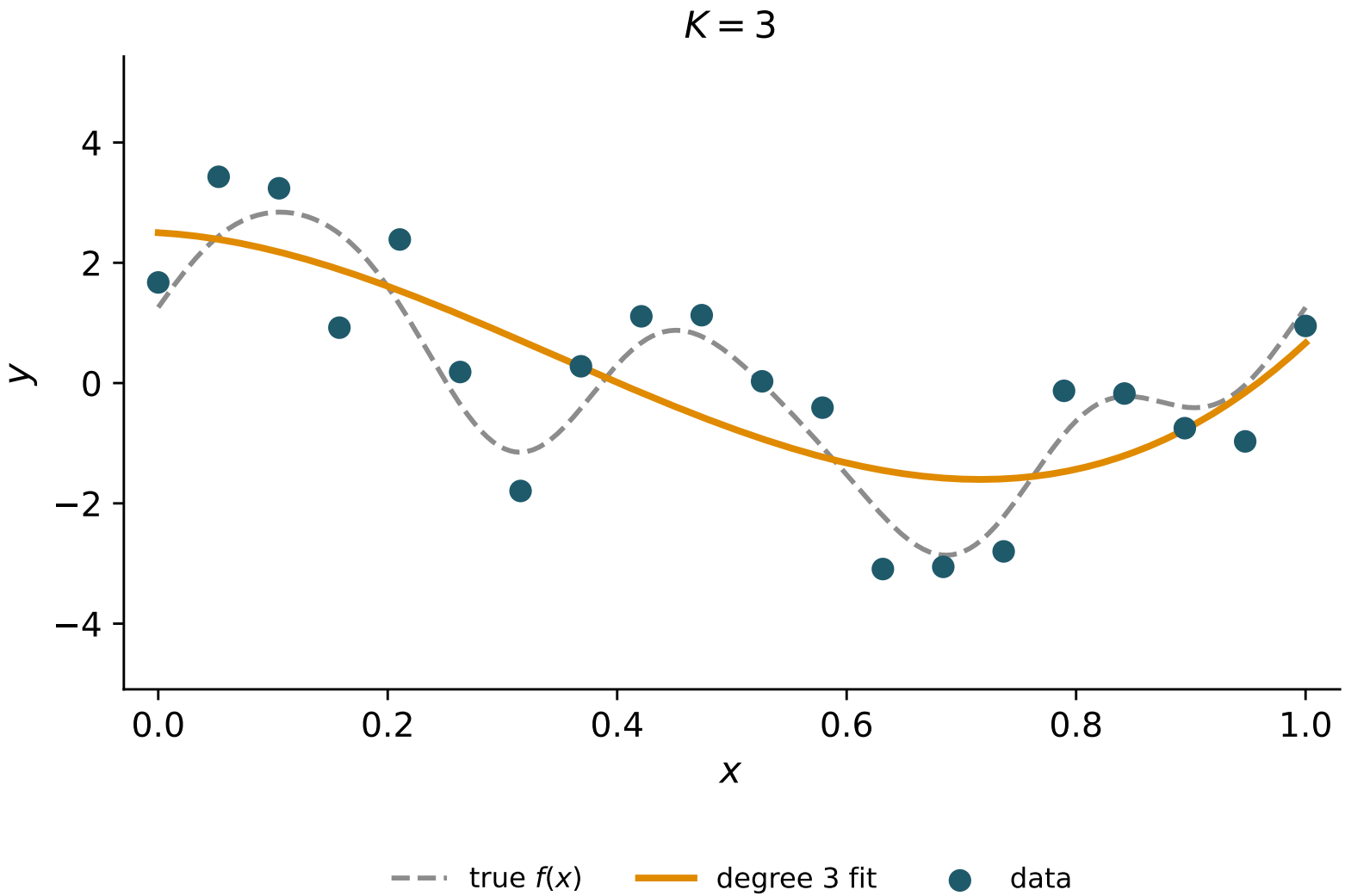
Example: polynomials fitted to a random function



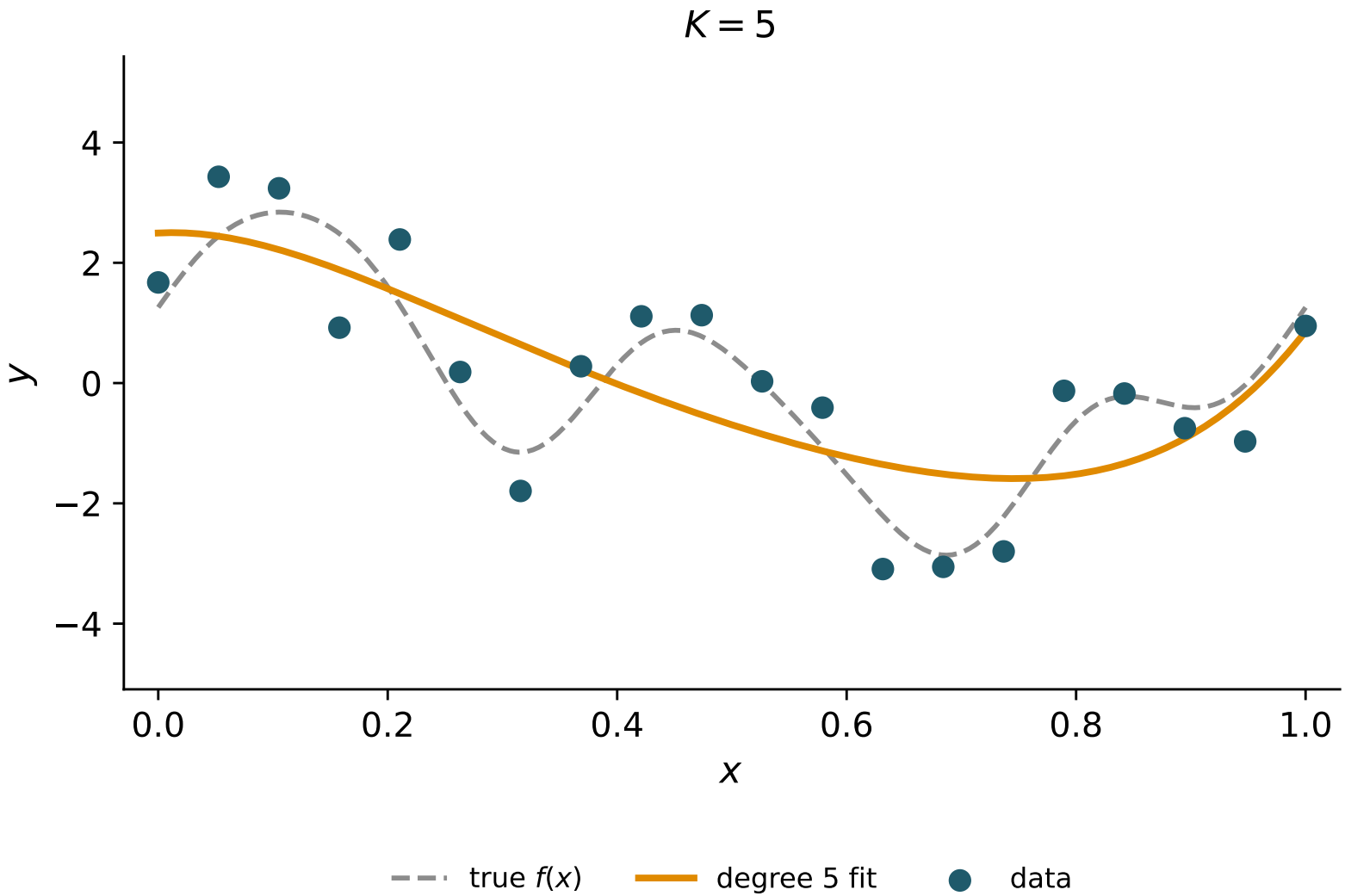
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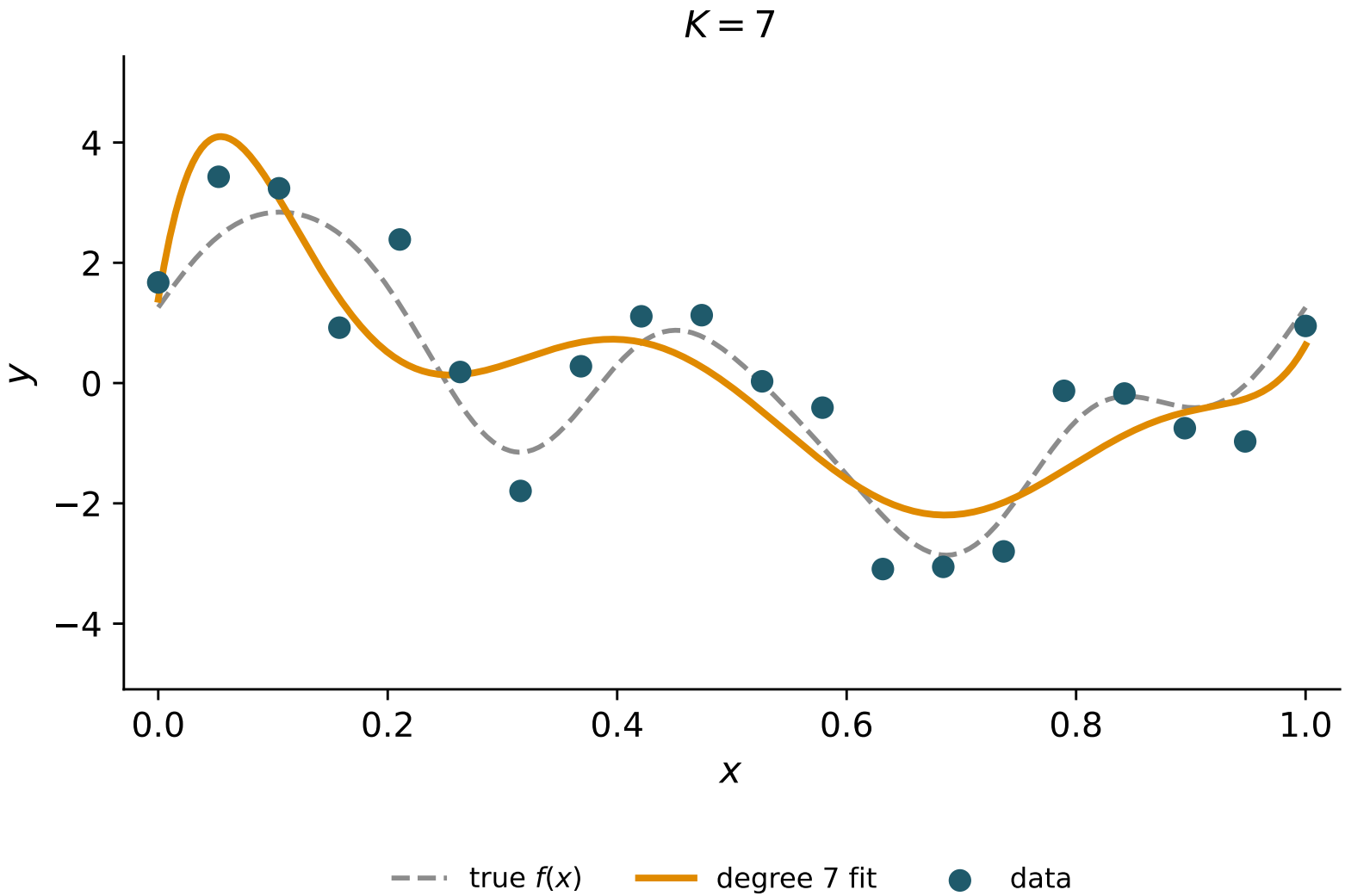
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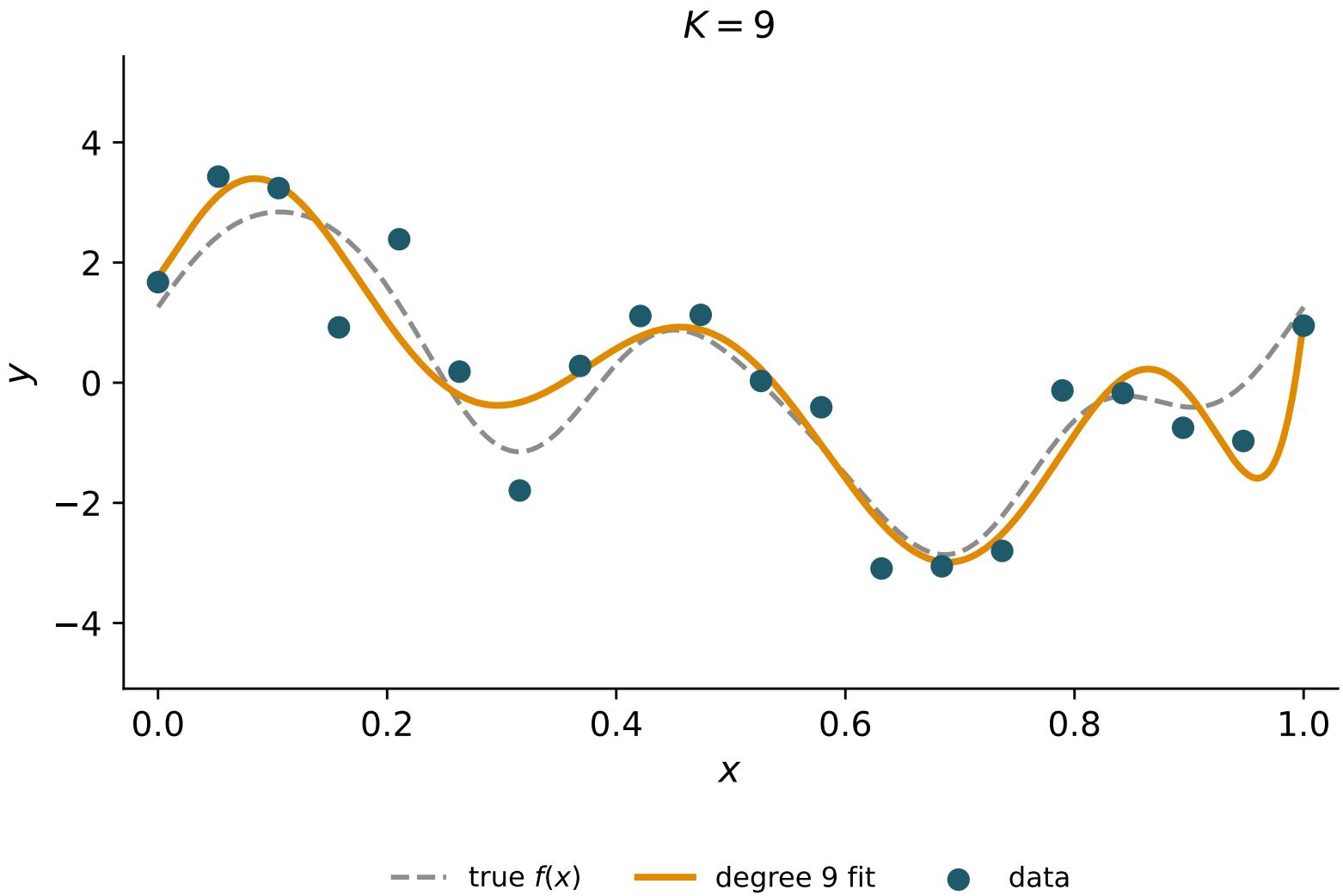
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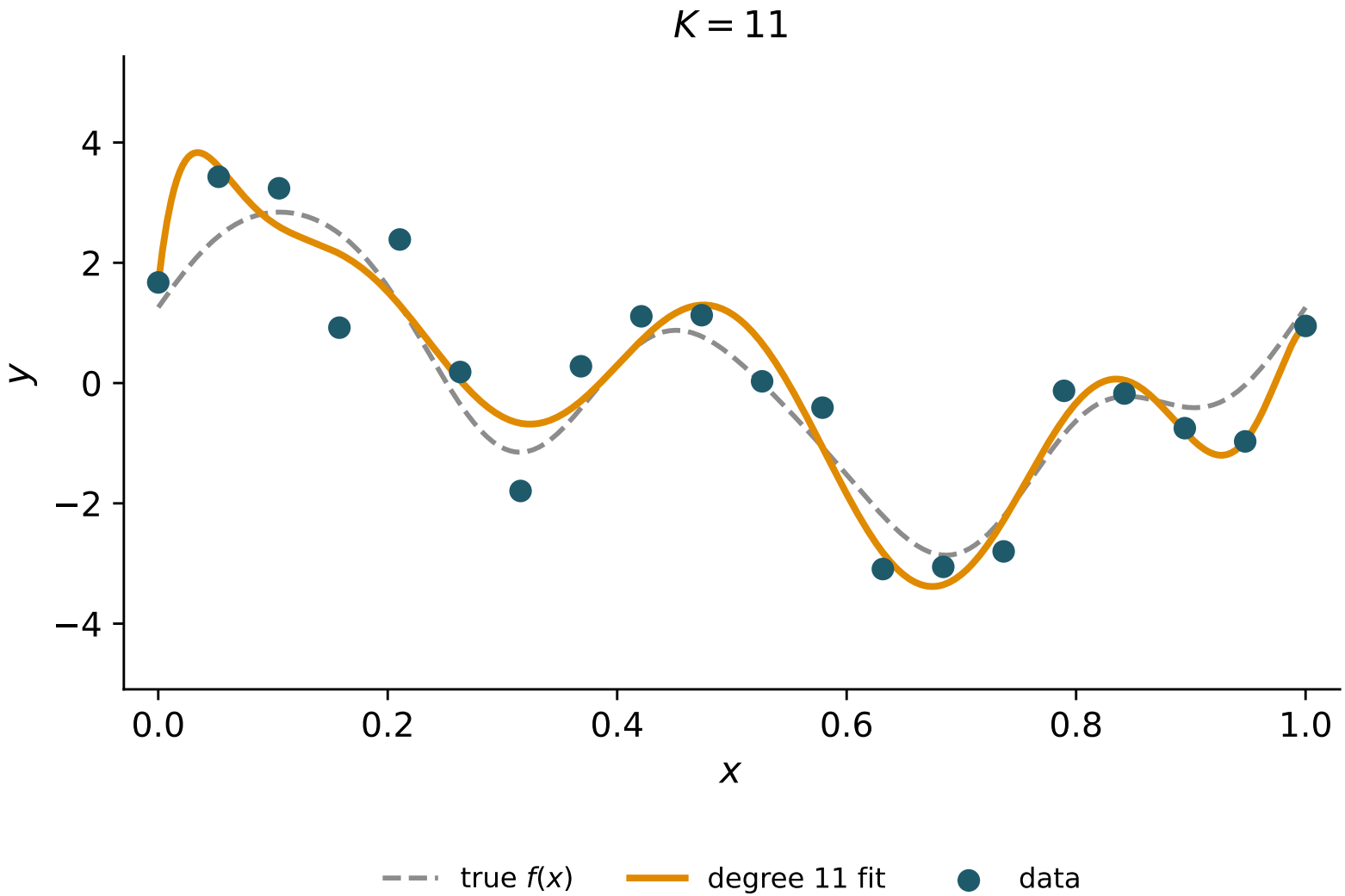
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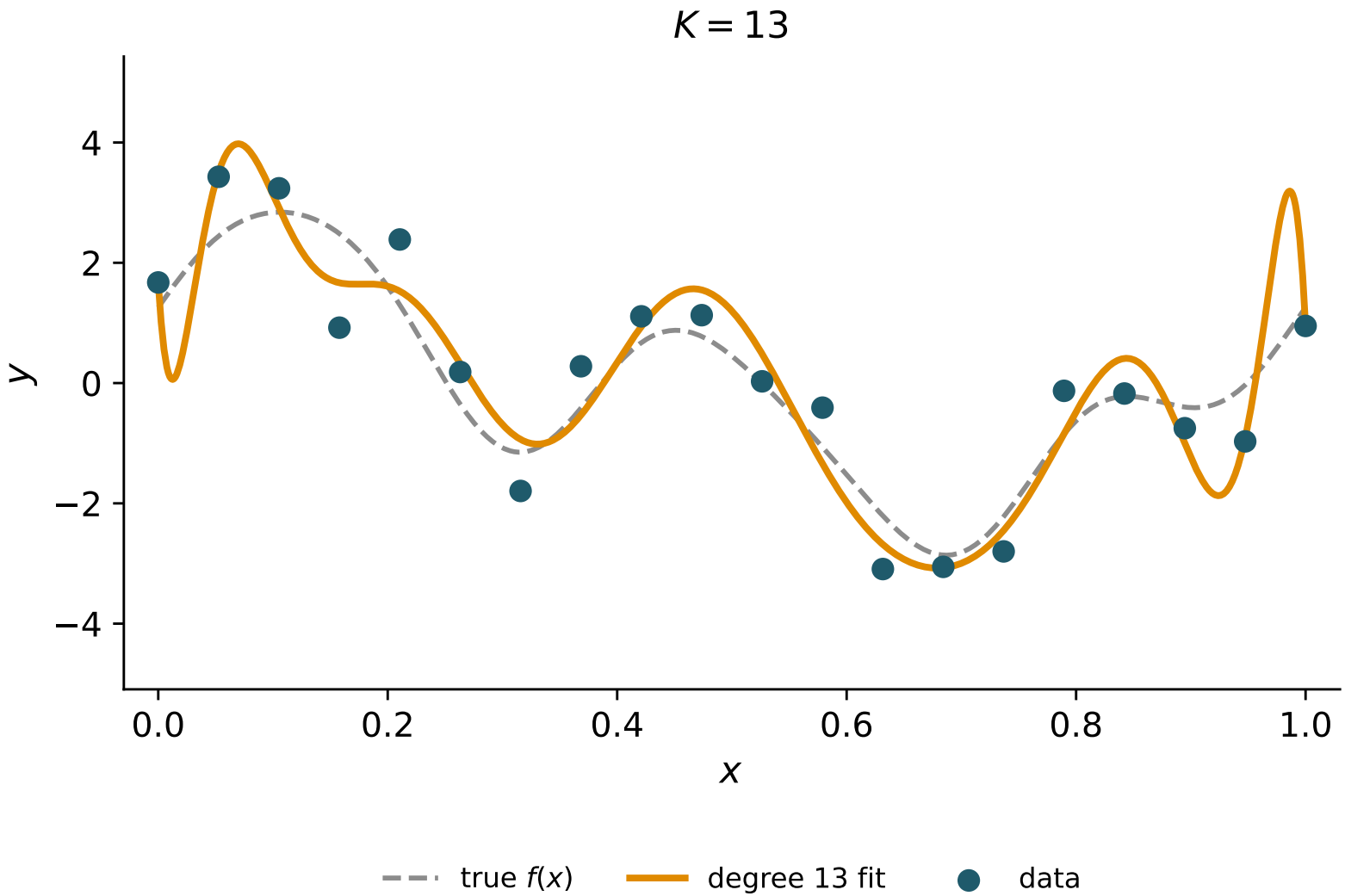
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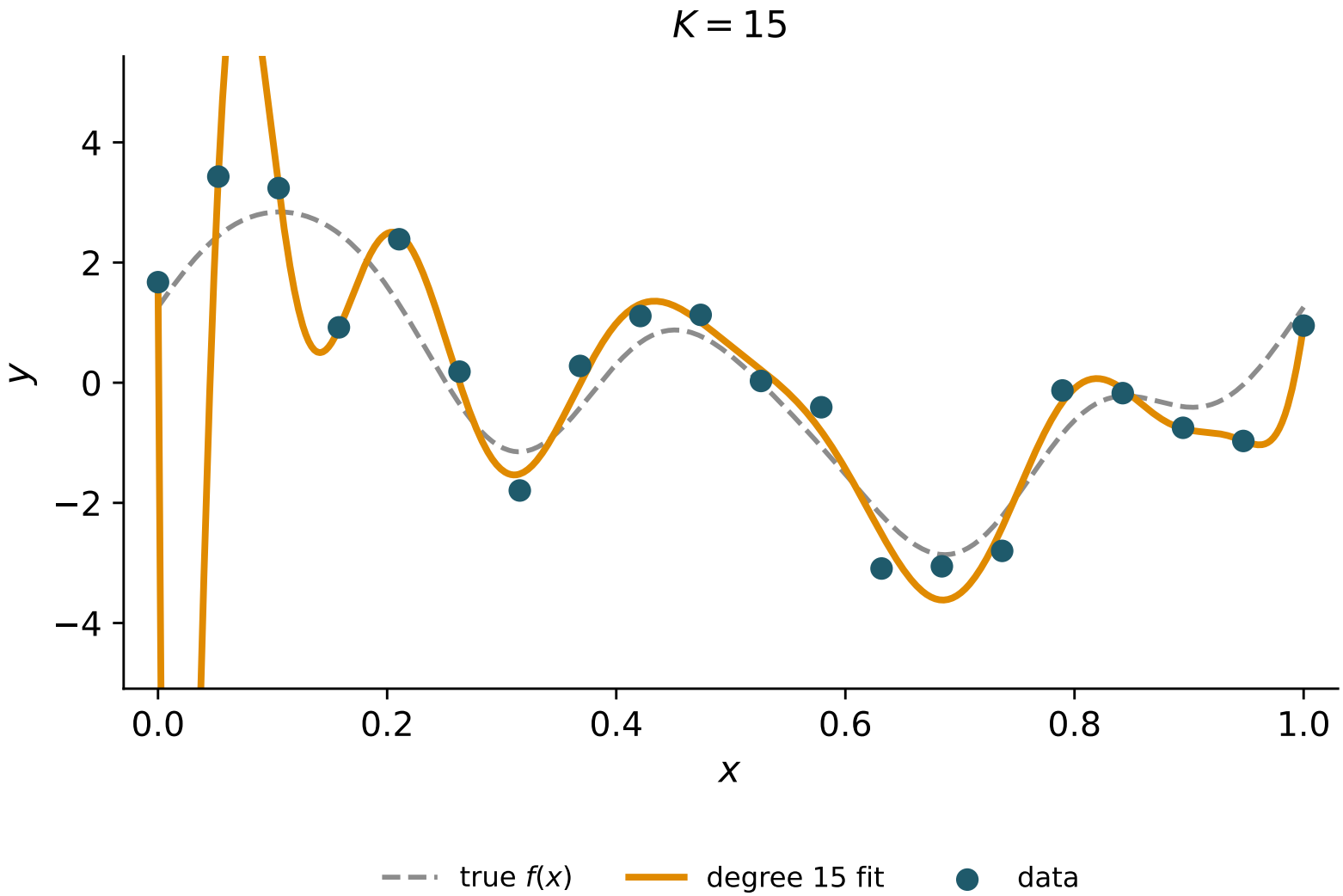
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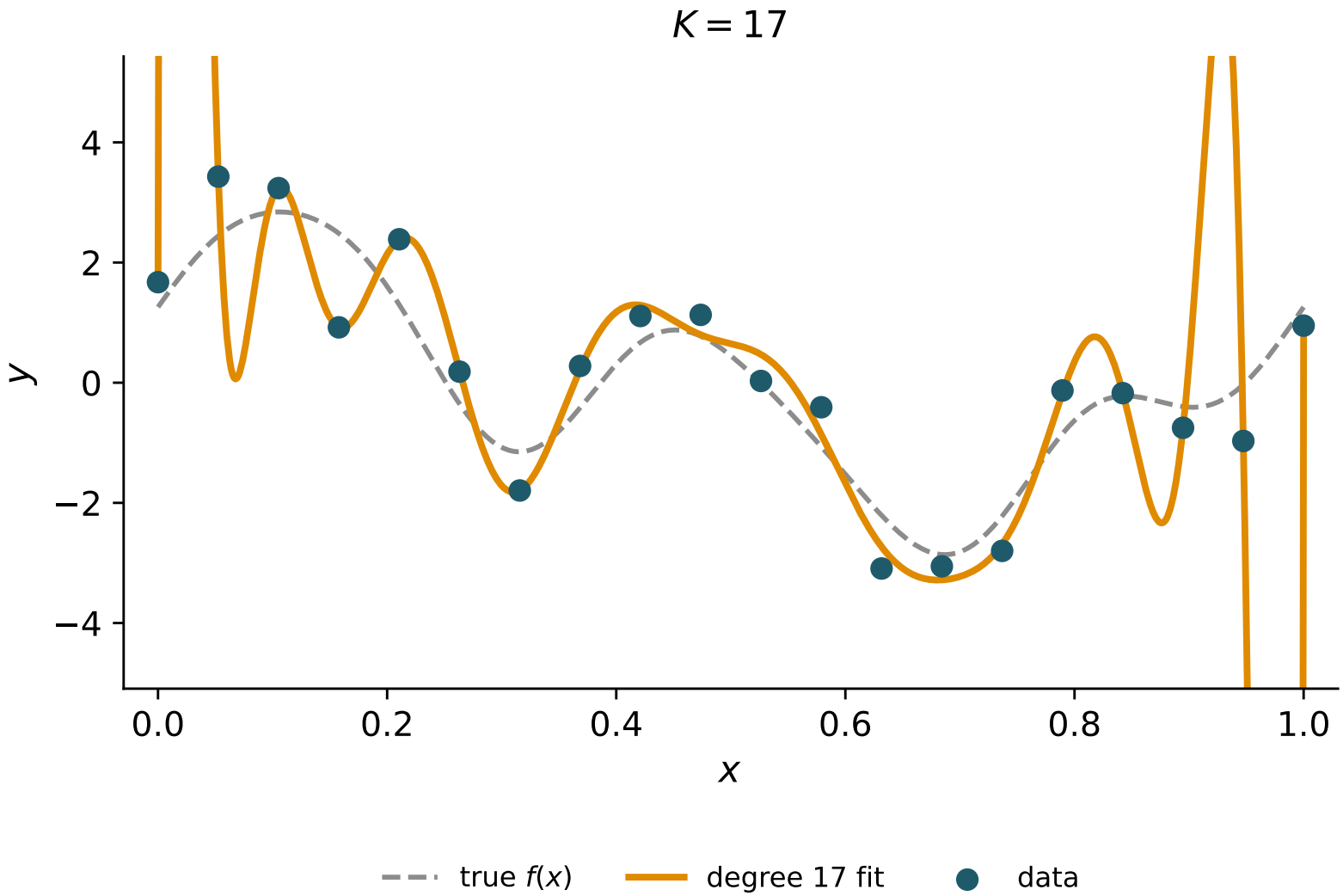
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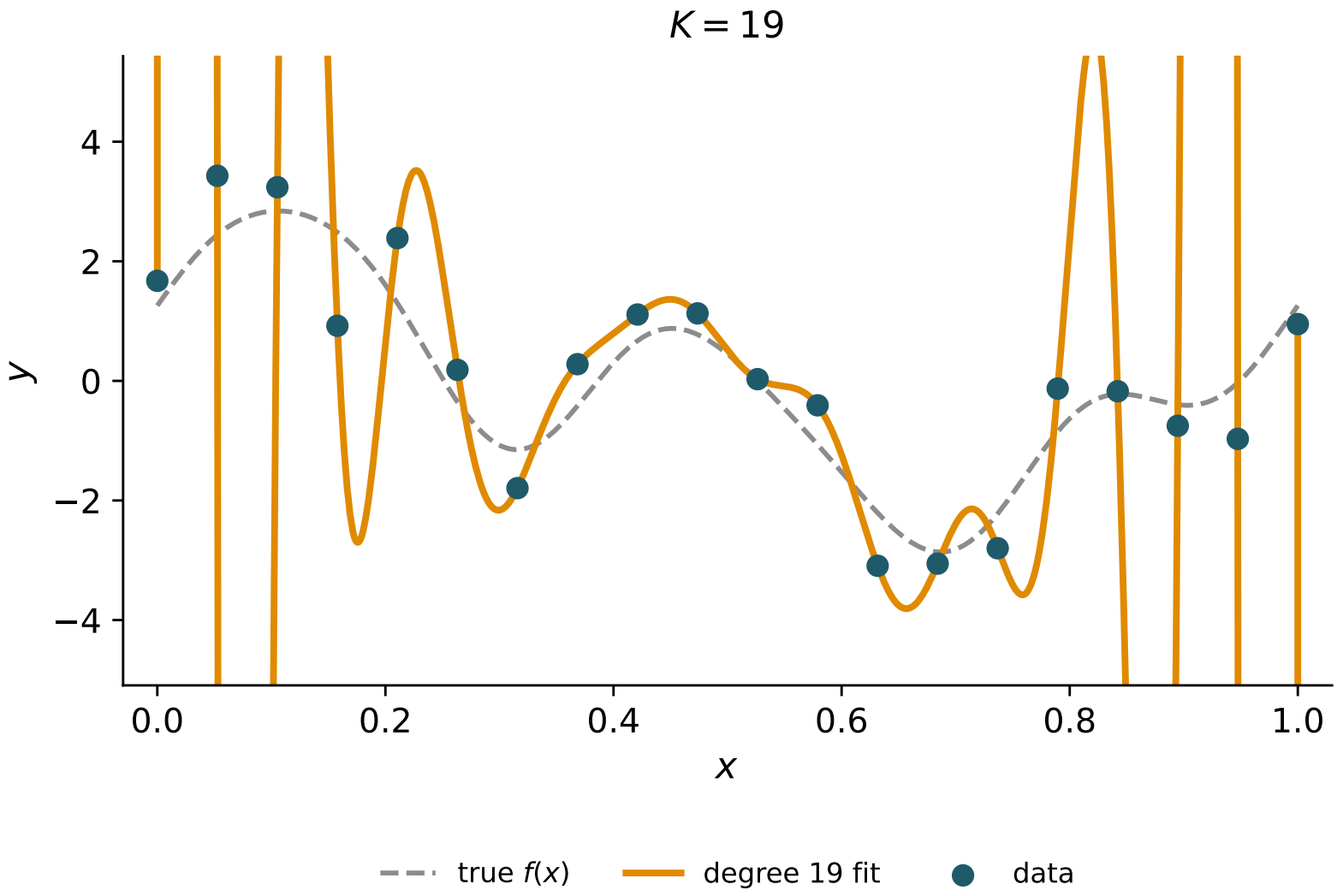
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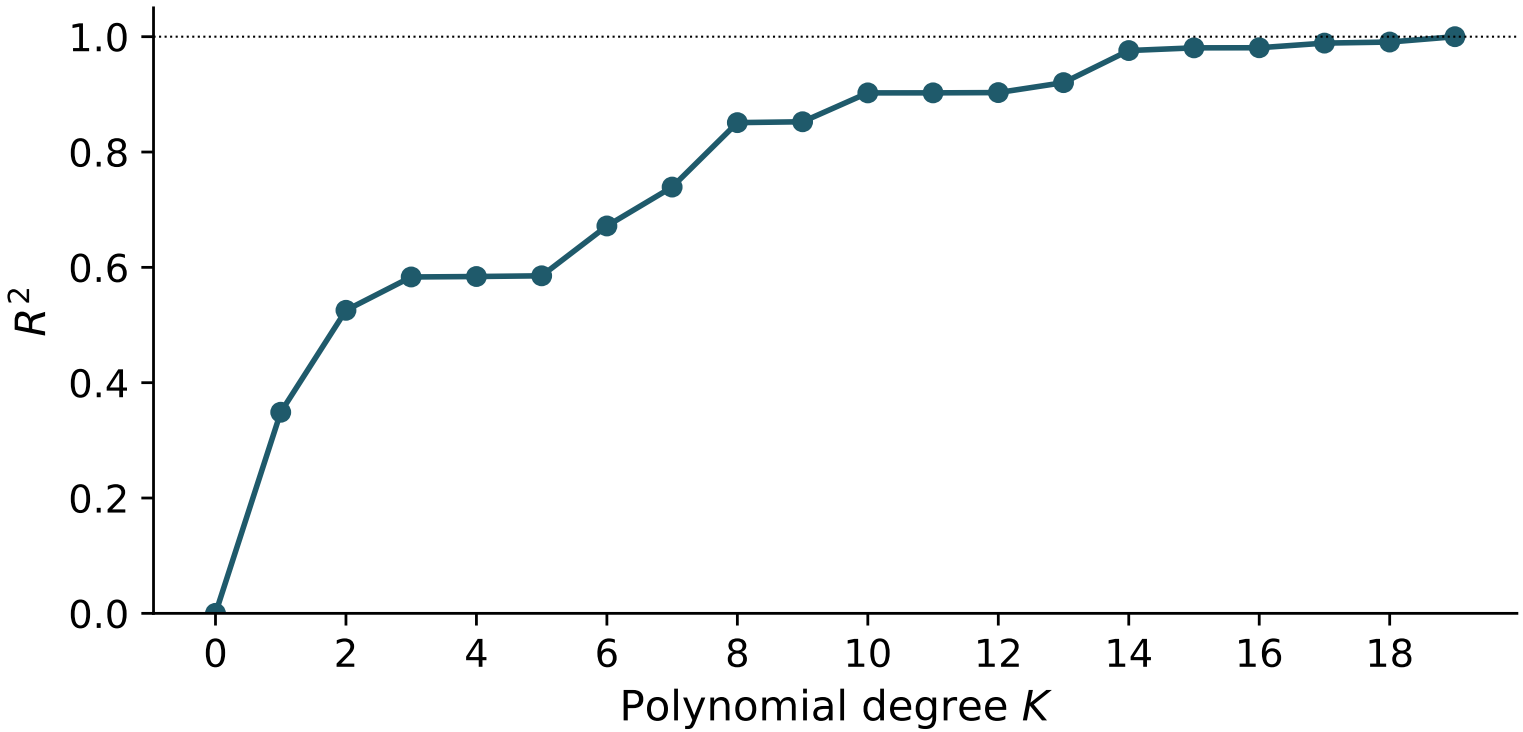


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R^2 as the degree increases

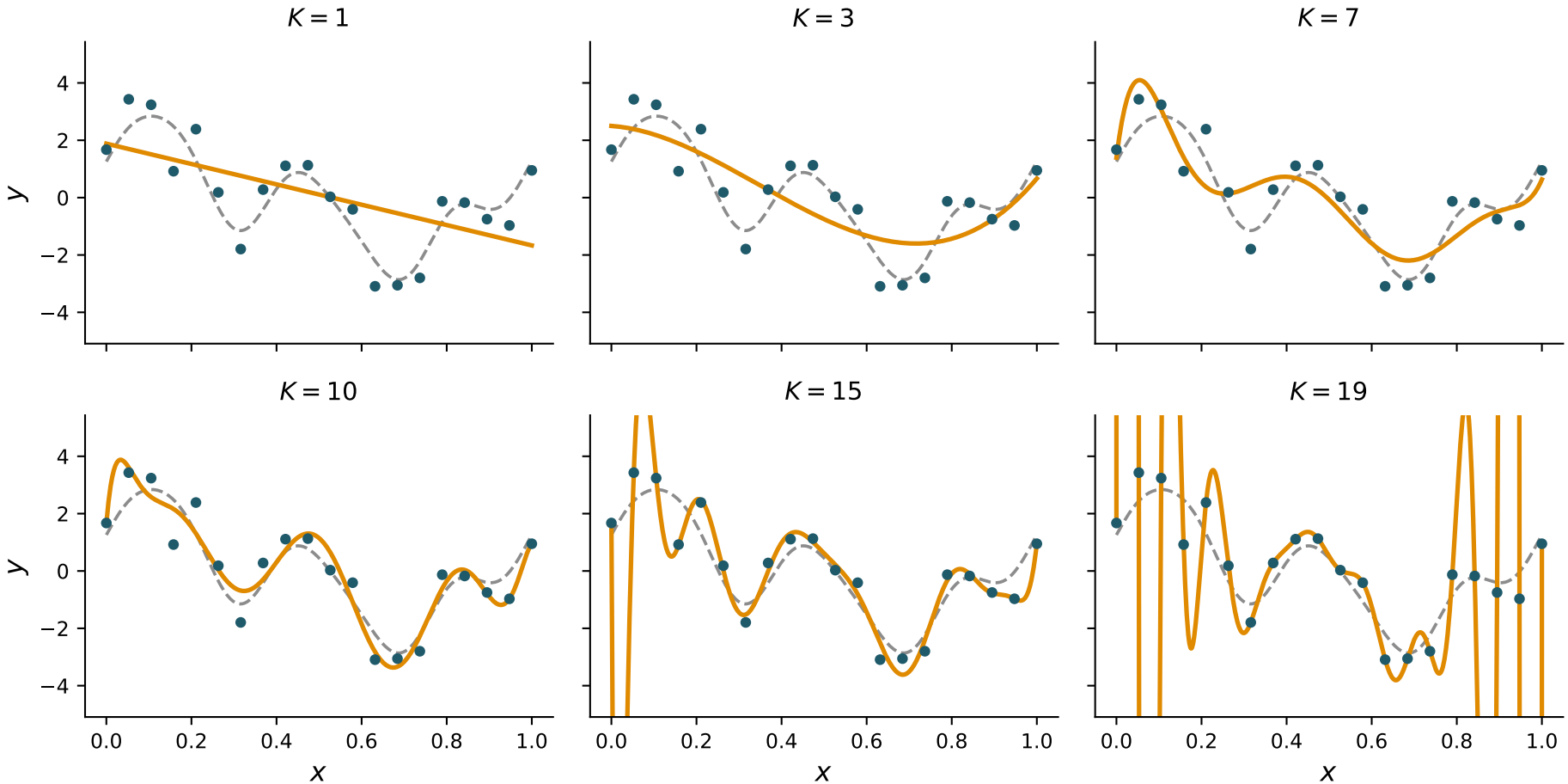
R^2 for the same 20 points and every degree $K = 0, 1, \ldots, 19$:



R^2 goes up with every coefficient we add and reaches 1 at $K = 19$, where the polynomial passes through all 20 points.

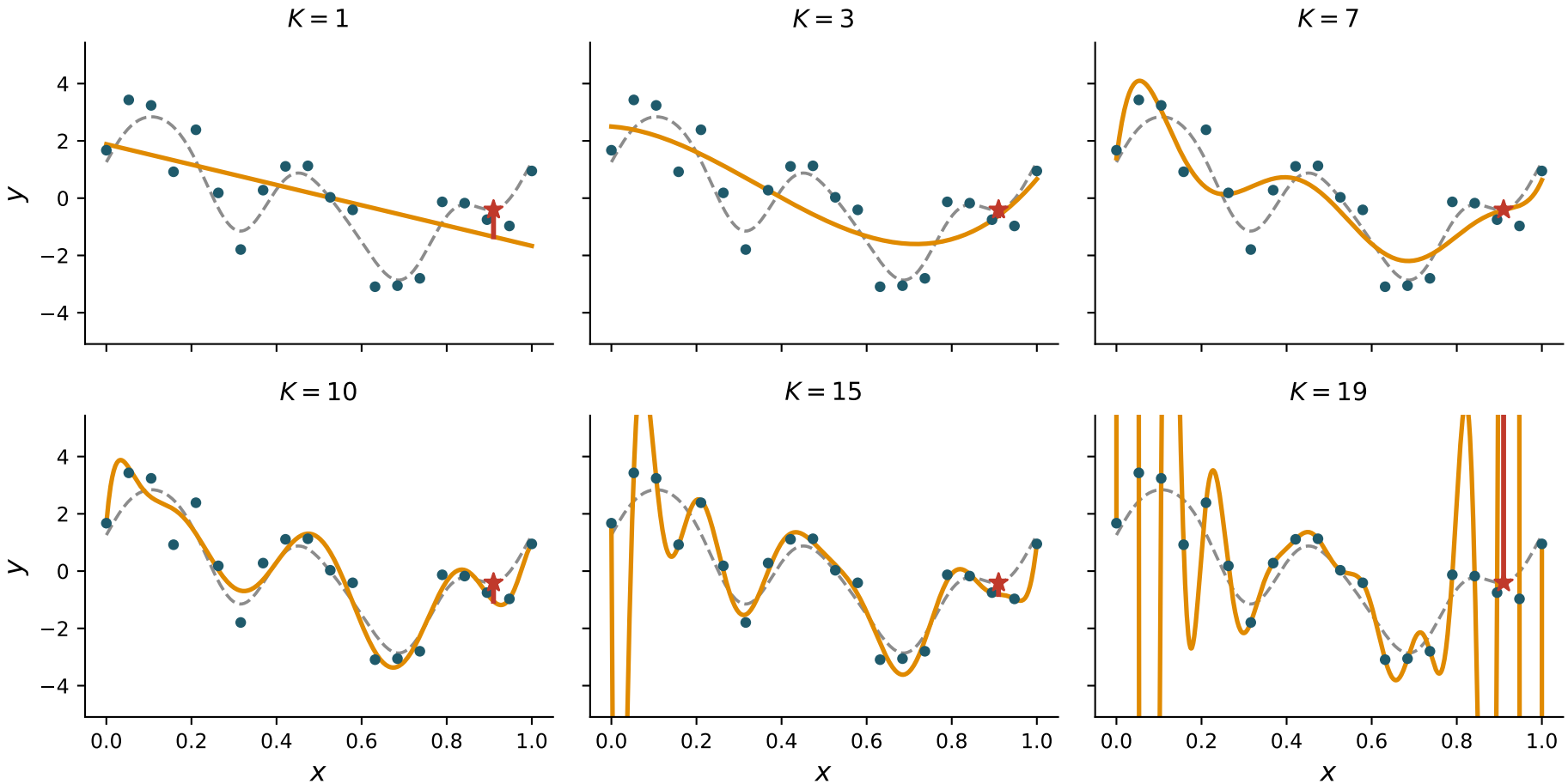
Predicting a new point

The true function (dashed), the 20 points and the fitted polynomials.



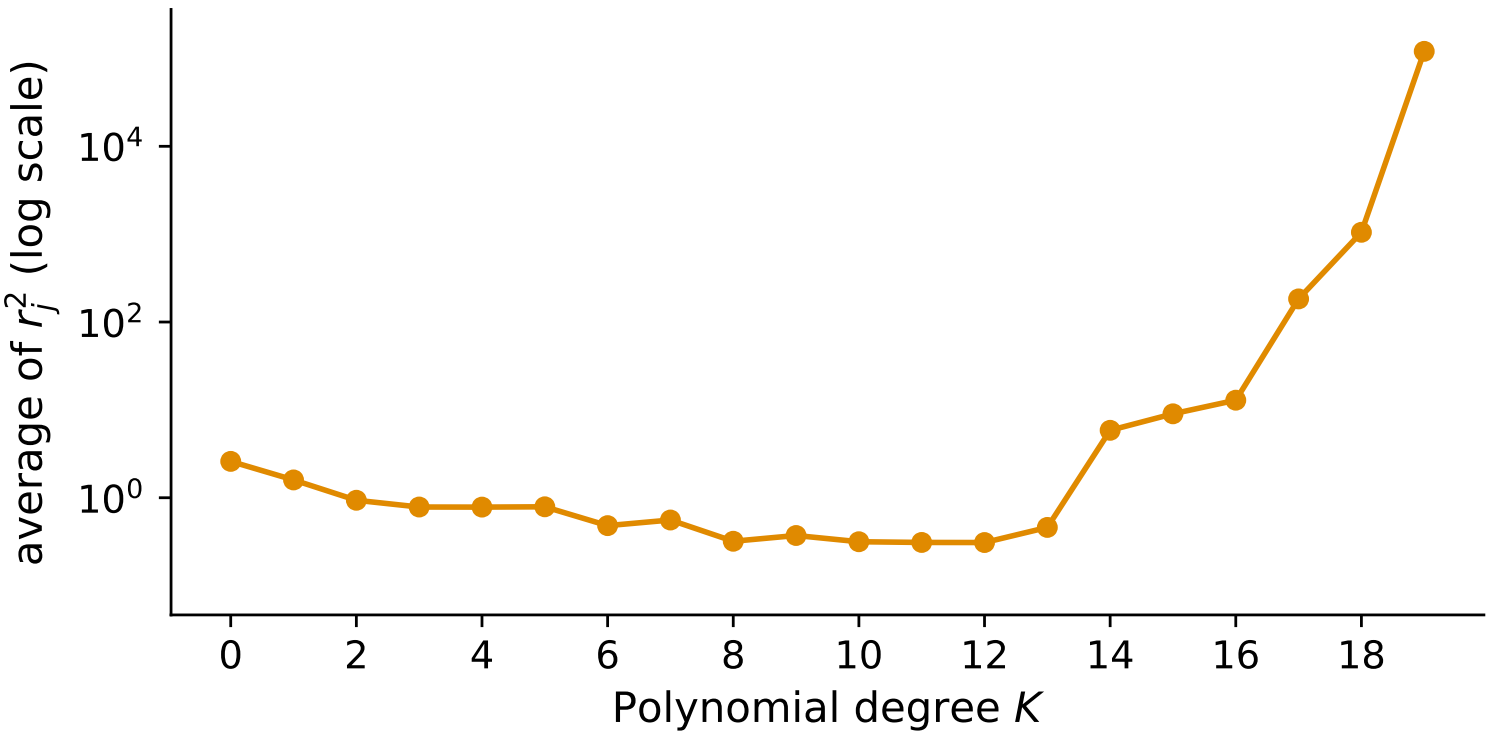
Predicting a new point

The true function (dashed), the 20 points and the fitted polynomials. A new point on the true function at $x = 0.91$ (star). Its residual $r = y - \hat{y}$ is the vertical distance from the fitted curve (red).



Average squared residual of the new points

Repeat this for 200 random points on the true function, with residuals r_j. For each degree K, the average of r_j^2:



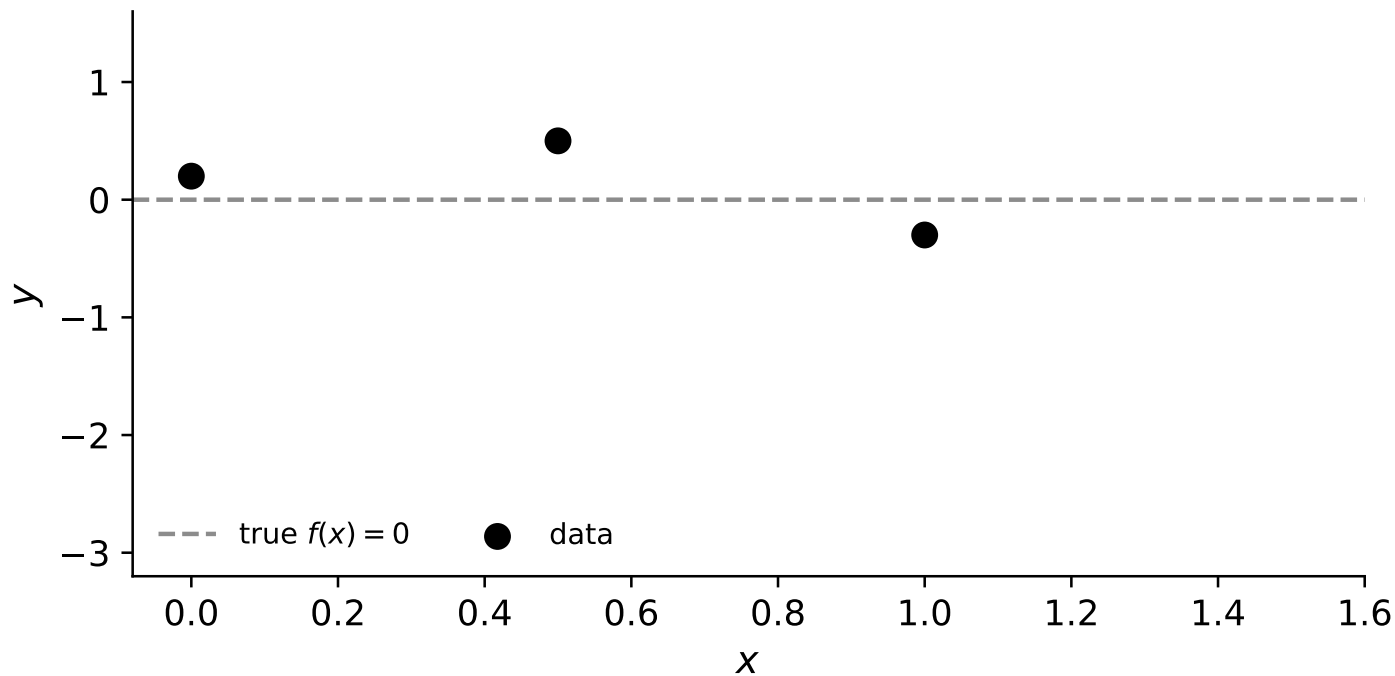
It is smallest around $K = 10$ to 12. For larger K it grows quickly, and at $K = 19$ it is about 120000.

Why the coefficients blow-up: minimal example

Consider the model

$$Y = \beta_0 + \beta_1 X + \beta_2 X^3 + \epsilon$$

Assume the data is generated with $\beta_0 = \beta_1 = \beta_2 = 0$, so there is no underlying relationship. Three data points, at $x = 0$, 0.5 and 1.

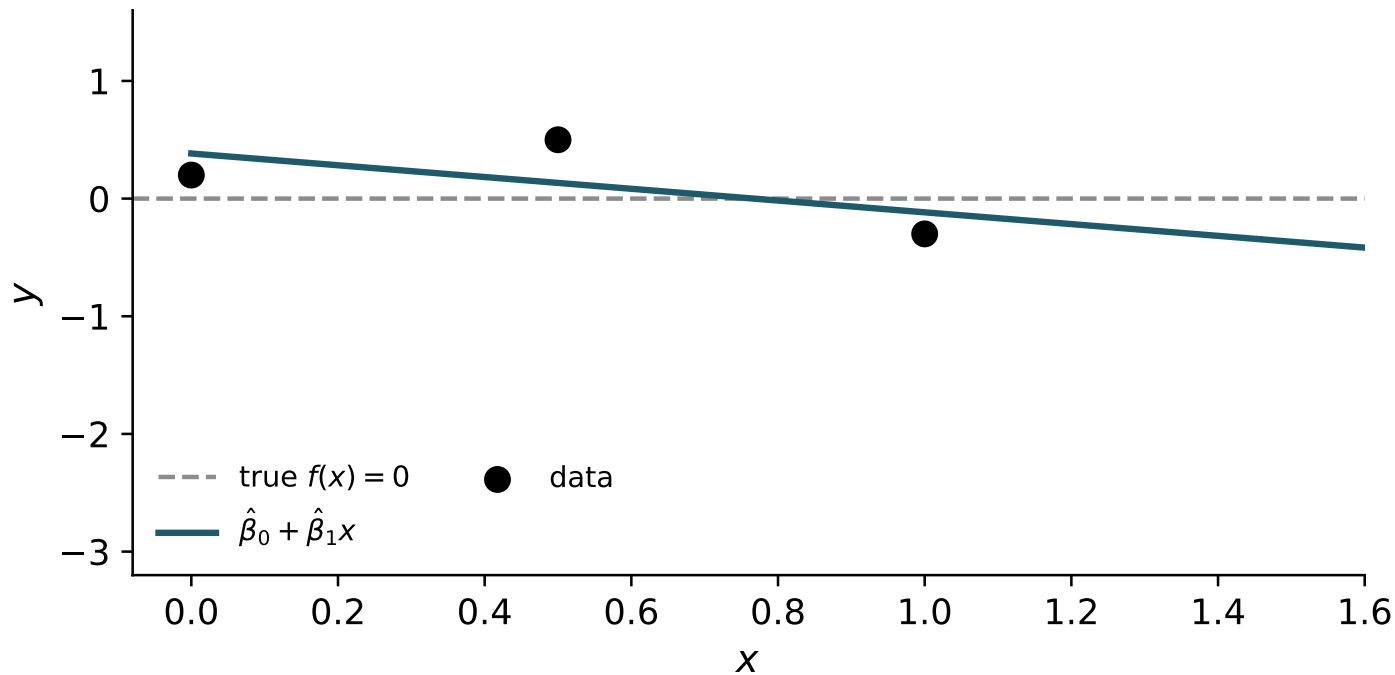


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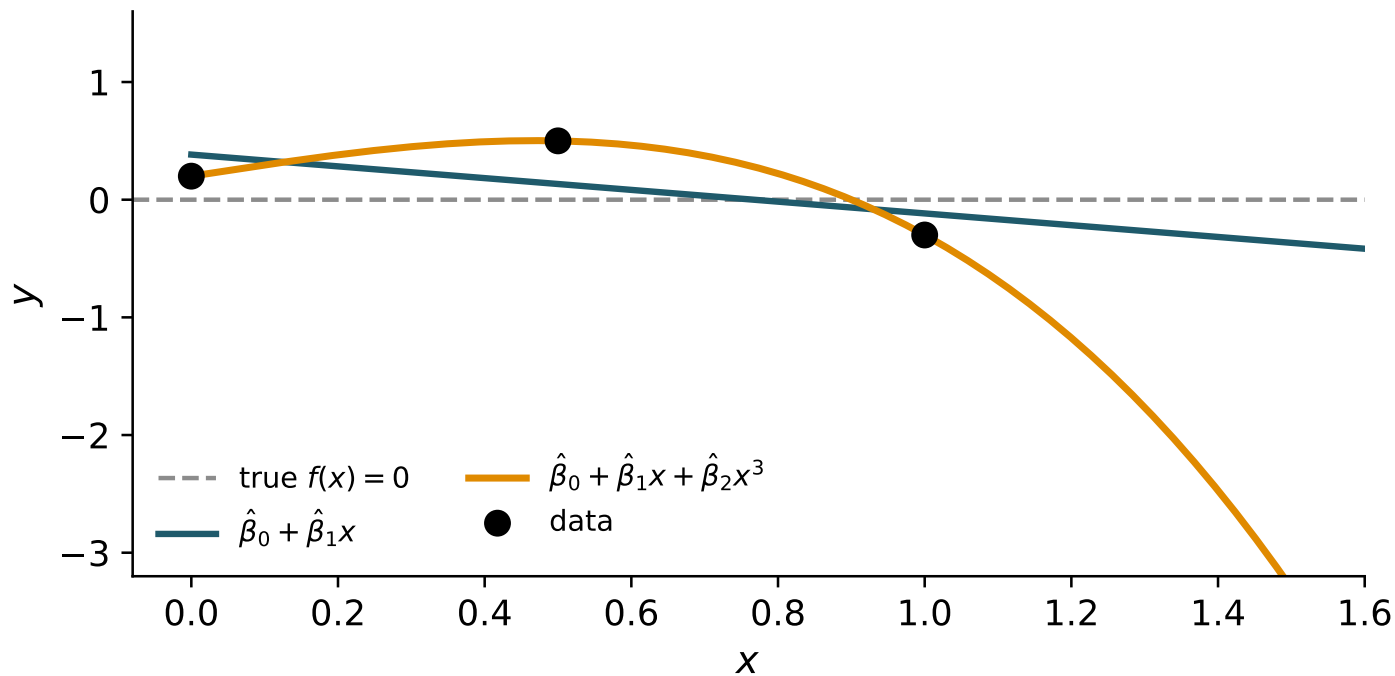
With only $\beta_0 + \beta_1 X$ (2 coefficients), the least-squares line cannot pass through all three points.

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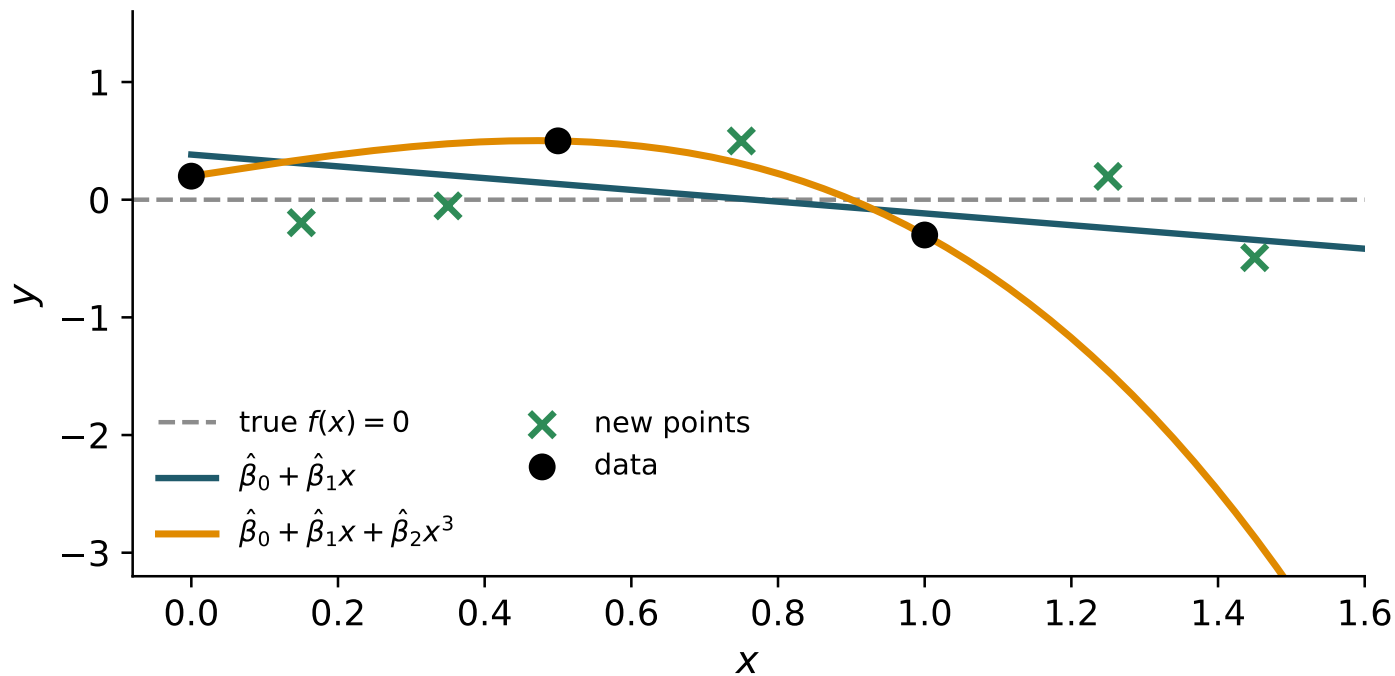


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Why are the coefficients so large?

The same model with all three coefficients 0 in reality, so $Y = \epsilon$. Three data points at $x = 0$, h and $2h$, with y values ϵ_0, ϵ_1 and ϵ_2. Three coefficients and three points, so the fitted curve passes through all of them:

$$\begin{aligned} \hat{\beta}_0 &= \epsilon_0 \\ \hat{\beta}_0 + \hat{\beta}_1 h + \hat{\beta}_2 h^3 &= \epsilon_1 \\ \hat{\beta}_0 + 2\hat{\beta}_1 h + 8\hat{\beta}_2 h^3 &= \epsilon_2 \end{aligned}$$

Question

Solve for $\hat{\beta}_2$. If ϵ_0, ϵ_1 and ϵ_2 are independent with variance σ_ϵ^2, what is $\mathrm{var}(\hat{\beta}_2)$? What happens to it as h gets smaller?

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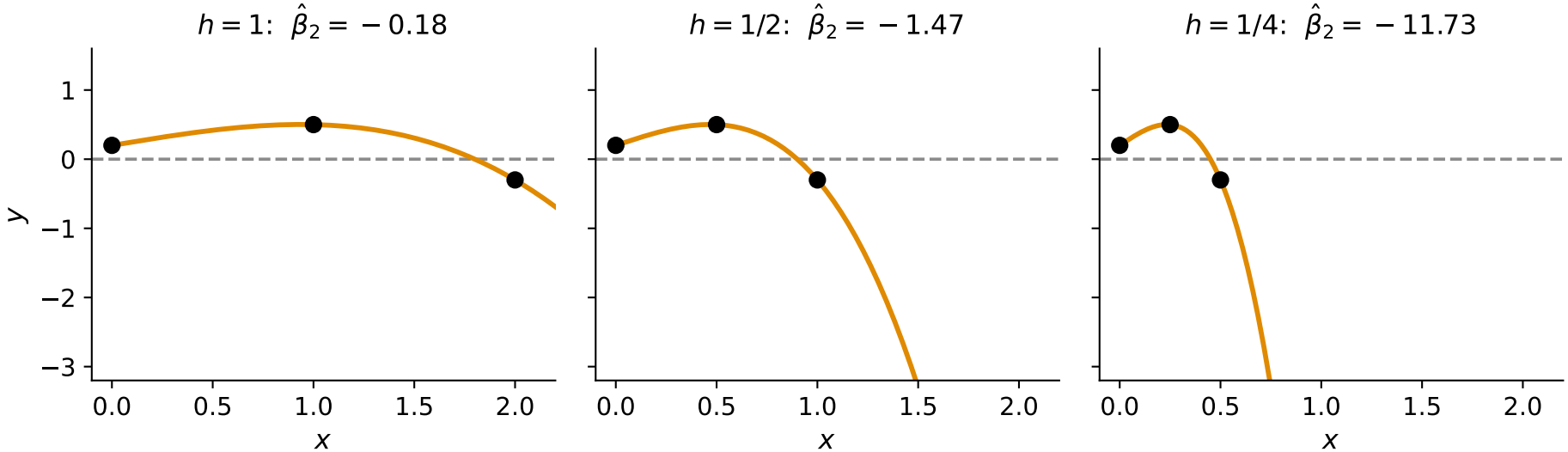
Put $\hat{\beta}_0 = \epsilon_0$ into the other two equations, then subtract twice the second from the third: $6\hat{\beta}_2 h^3 = \epsilon_2 - 2\epsilon_1 + \epsilon_0$, so

$$\hat{\beta}_2 = \frac{\epsilon_2 - 2\epsilon_1 + \epsilon_0}{6h^3}, \qquad \mathrm{var}(\hat{\beta}_2) = \frac{(1+4+1)\,\sigma_\epsilon^2}{36h^6} = \frac{\sigma_\epsilon^2}{6h^6}.$$

Halving h multiplies the standard deviation of $\hat{\beta}_2$ by $2^3 = 8$.

Why are the coefficients so large?

The same noise, $\epsilon_0 = 0.2$, $\epsilon_1 = 0.5$ and $\epsilon_2 = -0.3$, with the points closer and closer together:



Near 0, x^3 is tiny, so it takes a huge $\hat{\beta}_2$ to fit the noise. The degree-19 fit has the same problem with x^{19} and 20 points close together.